

Dauletkhan Assanov

 github.com/ab-dauletkhan
 linkedin.com/in/ab-dauletkhan

ab.dauletkhan@gmail.com
+7(705)700-2361
010000, Astana, Kazakhstan

EDUCATION

Bachelor of Science in Computer Science

August 2022 – June 2026

Nazarbayev University – Top 1 University in Kazakhstan

Astana, Kazakhstan

Coursework: Data Structures and Algorithms, Computer Organization, Operating Systems, Database Systems, Software Engineering, Software Architecture, Computer Networks, Artificial Intelligence, Linear Algebra, Discrete Mathematics

TECHNICAL SKILLS

Programming Languages: Go, Python, C++, Typescript, SQL

Databases: PostgreSQL, MongoDB, Redis

Tools & Platforms: Docker, Kubernetes, AWS S3, Apache Kafka, Git, GitHub Actions, GitLab CI, Linux

EXPERIENCE

Software Engineer

November 2024 – Present

Beeline – Leading telecom operator serving 11.6M clients in Kazakhstan

Astana, Kazakhstan

- Booster concurrent call capacity from **300** to **1,000+** sessions by driving migration from Asterisk to Kamailio + FreeSWITCH, delivering lower latency, higher reliability, and long-term scalability for telecom voice services
- Reduced infrastructure costs by **95%** by migrating 6,000+ lines of core API and billing code from Python to Go, improving request latency by **12%** and consolidating servers from **100** to **5**
- Reduced mean voice processing latency from **250ms** to **150ms** by implementing a Go microservice with concurrent STT/TTS session handling and smart request batching, ensuring smooth real-time customer interactions
- Cut customer service response time from **3 minutes** to **15 seconds** by linking AI bot services to billing systems via REST APIs for real-time balance and tariff inquiries
- Accelerated release cycles from **15 minutes** to **3 minutes** by streamlining the CI/CD pipeline with automated builds, tests, and deployments

Network Application Engineering Intern

August 2024 – October 2024

7Generation – An international specialist in telecom data analytics and monitoring software

Astana, Kazakhstan

- Enhanced Go-based packet processing module by implementing pattern-matching algorithms for L7 protocols (HTTP/2, DNS), improving anomaly detection accuracy by **15–25%**
- Optimized data processing pipeline using Go concurrency primitives for concurrent network flow analysis, achieving **30%** performance improvement in real-time monitoring systems
- Built and deployed network monitoring dashboard with Prometheus and Grafana, visualizing **10,000+** metrics per second for real-time network anomaly detection and system health monitoring

Software Engineer

February 2024 – August 2024

Global Optima Technologies – Local IT outsourcing by ex-Big Tech engineers

Astana, Kazakhstan

- Reduced average query execution time from **250ms** to **30ms** by adding strategic indexes and refactoring complex PostgreSQL queries, significantly enhancing application performance
- Reduced production incidents by **~20%** through expanding unit and integration test coverage (**40% → 88%**) for backend and frontend codebases, improving reliability and client trust
- Accelerated feature delivery **33%** through development of 15+ reusable React components and 3 Go microservices, reducing code duplication **40%** across projects

PROJECTS

Scalable, fault-tolerant S3-inspired object store

October 2024

- Achieved **11.4K** GET requests/sec (**~1.39 GB/s**) with average latency **72ms**, demonstrating exceptional read performance under high concurrent load
- Delivered **200** PUT requests/sec for **128KB** objects with p95 latency **1.8ms** at **99%** success rate, ensuring reliable write operations

CLI Image Processing Application

October 2024

- Achieved **~322 MP/s** processing throughput through parallel algorithm design and $O(n)$ complexity optimizations for large bitmap images.
- Reduced blur operation time from **85s** to **less than 1s** via separable convolution algorithm reducing computational complexity from $O(n^2)$ to $O(n)$.